

Production Deployment Guide - FAQ Chatbot

Follow this step-by-step guide to deploy the FastAPI FAQ Chatbot backend to a production Linux server (e.g., Ubuntu 20.04/22.04 VM).

Step 1: Install Server-Level Prerequisites

Log in to your production server terminal and install Python, PostgreSQL, and the `pgvector` extension:

```
# 1. Update the server package repository
sudo apt update && sudo apt upgrade -y

# 2. Install Python 3.10+, pip, and virtual environment utilities
sudo apt install python3 python3-pip python3-venv -y

# 3. Install PostgreSQL database server
sudo apt install postgresql postgresql-contrib -y

# 4. Install pgvector for your specific PostgreSQL version (assumes PostgreSQL 16)
sudo apt install postgresql-16-pgvector -y
```

Step 2: Upload Project Files

Copy your project code to the target directory on your server (e.g., `/var/www/faq\_chatbot/`).

Keep these files/directories:

- `app/` (all agents, routes, data, and database logic)
- `migrations/` (all SQL database migration files)
- `main.py`
- `run\_migrations.py`
- `requirements.txt`
- `.env`

Exclude these (do not copy):

- `.venv/`
 - `\_\_pycache\_\_/`
 - `logs/`
-

Step 3: Set Up Python Virtual Environment

Navigate to the project directory on your server and initialize the Python environment:

```
# 1. Create a clean virtual environment
python3 -m venv venv

# 2. Activate the virtual environment
source venv/bin/activate

# 3. Upgrade pip and install all required libraries
pip install -U pip
pip install -r requirements.txt
```

Step 4: Configure the `.env` File

Open the `.env` file on the server and update the configuration variables for the production environment:

- Database Connection: Set `DATABASE\_URL` and `CHECKPOINT\_DB\_URL` to point to your production PostgreSQL server.
 - LLM Provider: Configure the `LLM\_PROVIDER` (e.g., `groq`, `openai`, or `gemini`) and set the corresponding API keys.
 - Urls: Update `FRONTEND\_URL` and `BACKEND\_URL` with your public production domains.
-

Step 5: Run Database Migrations

Run the database migration script to enable `pgvector` and construct all the necessary tables (`users`, `sessions`, `messages`, `logs`, etc.):

```
python run_migrations.py
```

Step 6: Verify and Run the Application

Start the FastAPI server using Uvicorn. On the first startup, the application lifespan will:

1. Connect to the database and initialize the LangGraph checkpointer tables.
2. Download the semantic embedding model (if configured for online download).

3. Seed the FAQ questions and answers into the PostgreSQL vector database.

To launch the server with multiple parallel workers for production traffic:

```
uvicorn main:app --host 0.0.0.0 --port 8000 --workers 4
```

Step 7: Manage Server Lifespan with Systemd

To keep the application running continuously in the background and ensure it automatically restarts if the server reboots or crashes, create a systemd service:

1. Create the service file:

```
sudo nano /etc/systemd/system/faq-chatbot.service
```

2. Paste the following configuration (replace path placeholders with your actual directory path):

```
[Unit]
```

```
Description=FastAPI FAQ Chatbot Service
```

```
After=network.target
```

```
[Service]
```

```
User=www-data
```

```
WorkingDirectory=/var/www/faq_chatbot
```

```
ExecStart=/var/www/faq_chatbot/venv/bin/uvicorn main:app --host 127.0.0.1 --port 8000
```

```
--workers 4
```

```
Restart=always
```

```
[Install]
```

```
WantedBy=multi-user.target
```

3. Enable and start the background service:

```
sudo systemctl daemon-reload
```

```
sudo systemctl enable faq-chatbot
```

```
sudo systemctl start faq-chatbot
```

Step 8: Configure Nginx as a Reverse Proxy with SSL (HTTPS)

Serve the chatbot backend securely over HTTPS:

1. Install Nginx and Certbot:

```
sudo apt install nginx certbot python3-certbot-nginx -y
```

2. Open Nginx configuration:

```
sudo nano /etc/nginx/sites-available/faq_chatbot
```

3. Add the server proxy block:

```
server {  
    server_name yourdomain.com;  
  
    location / {  
        proxy_pass http://127.0.0.1:8000;  
        proxy_set_header Host $host;  
        proxy_set_header X-Real-IP $remote_addr;  
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;  
        proxy_set_header X-Forwarded-Proto $scheme;  
    }  
}
```

4. Enable the site and restart Nginx:

```
sudo ln -s /etc/nginx/sites-available/faq_chatbot /etc/nginx/sites-enabled/  
sudo systemctl restart nginx
```

5. Request and install an SSL certificate:

```
sudo certbot --nginx -d yourdomain.com
```